

The Job Churn Ladder

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Introduction

- Excess turnover, aka "churn" is the separations or hires in excess of the net employment change of the firm.
- Churn arises from poaching in search models feat. a job ladder (Burdett & Mortensen (1998), Postel-Vinay and Robin (2002), Moscarini & Postel-Vinay (2013), etc.)
- Implication of job ladder models is monotone ladder in churn (and wage) and churn/wage correlation at firm and worker level (Bagger & Lentz(2018))

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 - **Separations to nonemployment**, driven by worker heterogeneity, is key to matching the data.
 - **Endogenous sorting** accounts for about half of the cross-sectional dispersion of firms.

Takeaway

- Unobservable worker heterogeneity in attachment matters for understanding labor market dynamics, flows. Hall and Kudlyak (2019); Gregory et al. (2021), Ahn et al. (2023)
- Growing literature linking firm characteristics, dynamics with worker outcomes. Tanaka et al (2023)
- Worker separation heterogeneity is also key to understanding the firm churn distribution.
 - Composition of high and low separation workers changes over the wage distribution.
- Role of frictions in generating dispersion (Wages: Hornstein, Krusell & Violante (2011) and Ortego-Marti (2016))

Data

- We use LEHD matched employer-employee data for 17 states from 1998-2013
- 15% random population sample, dropping firms < 10 workers
- Job-to-Job transitions: overlapping quarters of employment, change in dominant job
- 1.2 million job transitions

Measurement

- Measure annual flows of firm offset 1 quarter from the transition period

- Define $H_{j,t} = \frac{\sum_{k=1}^4 \mathbb{H}_{j,t-k}}{\frac{1}{2}(L_{j,t} + L_{j,t-4})}$ and $S_{j,t} = \frac{\sum_{k=1}^4 S_{j,t-k}}{\frac{1}{2}(L_{j,t} + L_{j,t-4})}$

- Net Employment Growth Rate:

$$\Delta L_{j,t} = H_{j,t} - S_{j,t}$$

- Excess Accession (Churn) Rate:

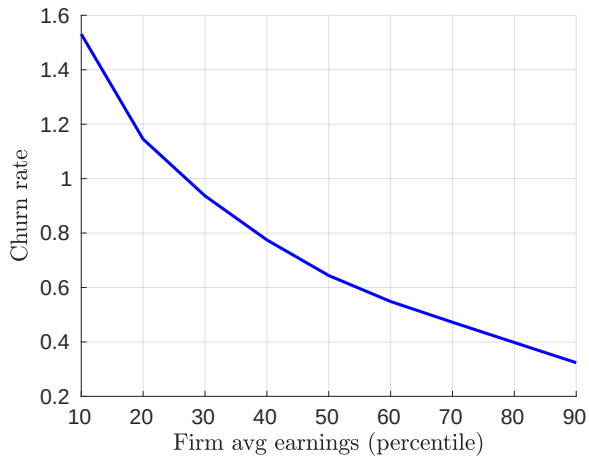
$$EAR_{j,t} = H_{j,t} - \max(\Delta L_{j,t}, 0) = \min(H_{j,t}, S_{j,t})$$

- Firms' Average Wage

$EW_{j,t}$: mean of employees' quarterly earnings

Churn and Average Wages

Firm Distribution



Measuring the job ladder

Excess Accessions Rate (Churn)

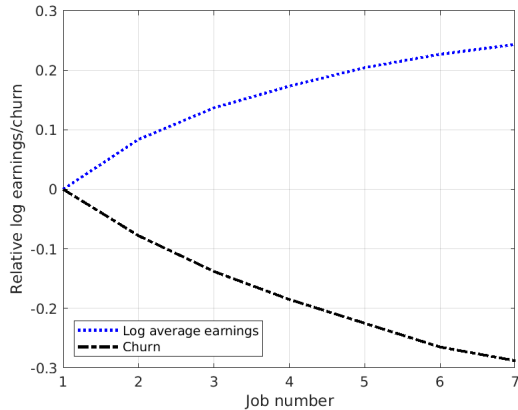
- Index jobs i within continuous employment spell j

-

$$EAR_{ij} = \beta_0 + \sum_{i=1}^N \beta_i \mathbb{I}_{job_i} + \gamma_j + \varepsilon$$

- Control for heterogeneity across workers and across employment spells with γ_j .
- We can sum coefficients and plot the change at each job index.

The Job Ladder



Estimated with emp. spell FEs

► Regression Specification

The Churn Ladder in a Search Model

ENVIRONMENT Burdett Mortensen (1998).

- Measure 1 of workers, pos. measure of firms of productivity z
- Random search in unemployment λ_u and on the job λ_e
- Exogenous separation into nonemployment δ
- Wage posting (constant, with commitment) and unemployment benefit b
- Offer distribution $F(w)$ results in realized wage distribution $G(w)$
- Firm size, conditional on w is $L(w)$

Introducing Worker, Firm Heterogeneity

- Permanent types, measure p has separation probability $s \in (0, \bar{\delta})$
- Types are unobservable to firm
- Reallocation shock λ_r to anywhere in the F distribution
- Firm-level separation rate elasticity with wage: $e^{\varepsilon_{\delta} w}$

The Churn Ladder in a Search Model

Flow balance without Heterogeneity

The stationary distribution implies a flow balance between inflows and outflows

$$\underbrace{(\delta + \lambda_e(1 - F(w)))L(w)}_{\text{outflows}} = \underbrace{\lambda_u u + \lambda_e G(w)(1 - u)}_{\text{inflows}}$$

- All flows are gross flows
- Separation rate is clearly decreasing in wages because $1 - F(w)$
- Hire rate increases in wages but more slowly because $G < F$

The churn rate per worker is

$$c(w) = \delta + \lambda_e(1 - F(w))$$

The Churn Ladder in a Search Model

Adding Heterogeneity

Type is proportion p with separation rate $s \in (0, \bar{\delta})$

$$c(w) = \underbrace{e^{\varepsilon_{\delta} w} \int_0^{\bar{\delta}} s p d(s|w)}_{\delta(w,s)} + \lambda_r + \lambda_e (1 - F(w))$$

$$\int_{\underline{w}}^w \int_0^{\bar{\delta}} (\delta(w, s) + \lambda_r (1 - F(w)) + \lambda_e (1 - F(w))) (1 - u(s)) h(w, s) ds dw =$$
$$\lambda_u u F(w) + F(w) \int_w^{\bar{w}} \int_0^{\bar{\delta}} \lambda_r (1 - u(s)) h(w, s) ds dw$$

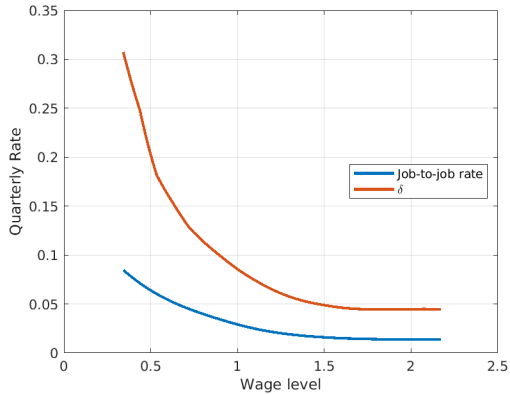
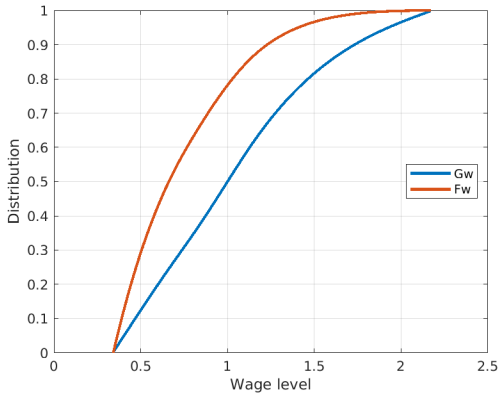
We only observe “ $G(w)$ ” and $\delta(w)$, but underlying is a joint distribution $H(w, s)$. We can use structure of the model to back out these joint distributions.

Calibration

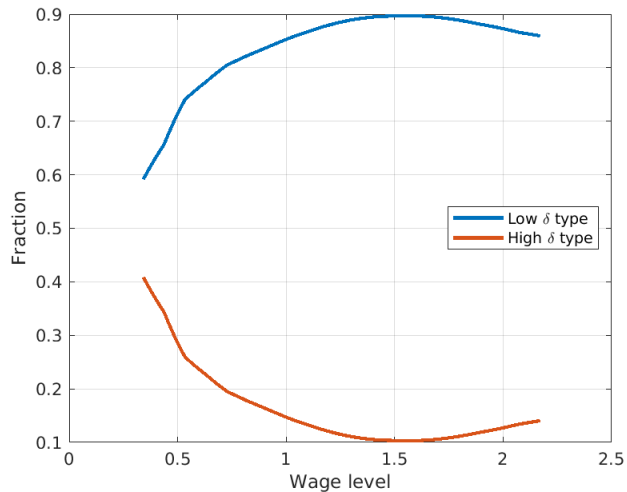
Take normalized $G(w)$ distribution from data - emp. weighted firm avg wage
Discretize wages and 2 types, proportion p_1 and p_2 with s_1 and s_2

Parameter	Value	Target
s_1	0.01	$\bar{\delta}(w) = .07$
s_2	0.45	
λ_e	0.022	Quarterly EE rate 0.035 (J2J)
λ_u	0.084	Quarterly NE rate 0.084
λ_r	0.014	est. - slope of worker ladder
$\varepsilon_{\delta w}$	-0.529	est. - firm churn/wage dist.
p_1	0.55	est. - $\delta(w) / J2J(w)$
p_2	0.45	

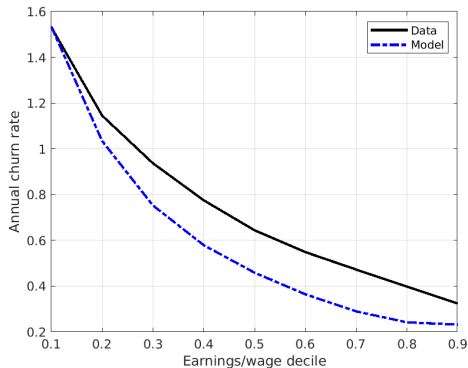
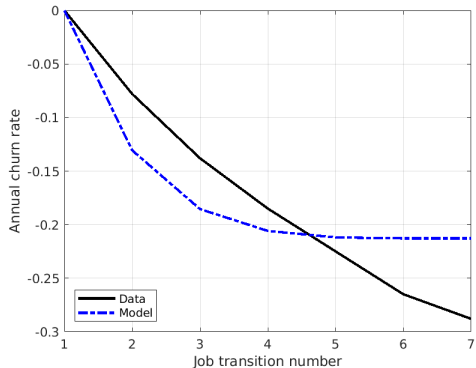
Calibrated churn ladder: Heterogeneous Types



Calibrated churn ladder: Sorting over the Job Ladder

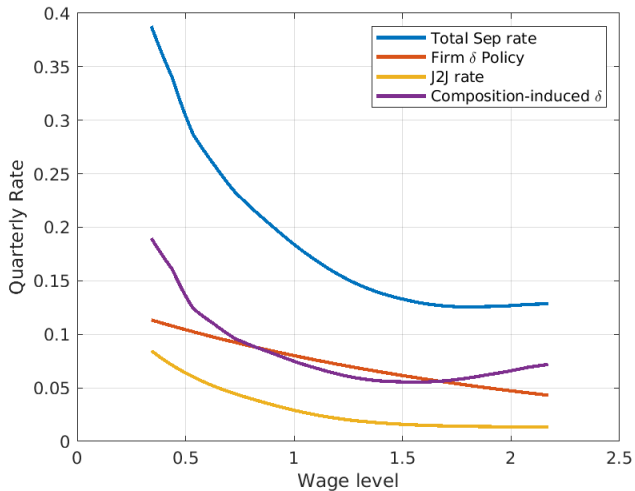


Calibrated churn ladder: With Worker Heterogeneity

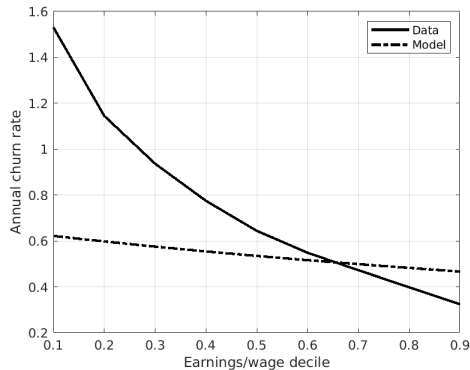
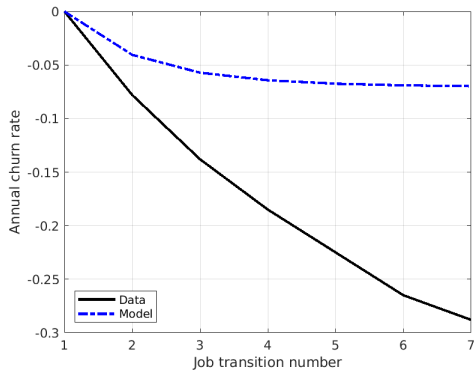


► Const. Fraction of Seps.

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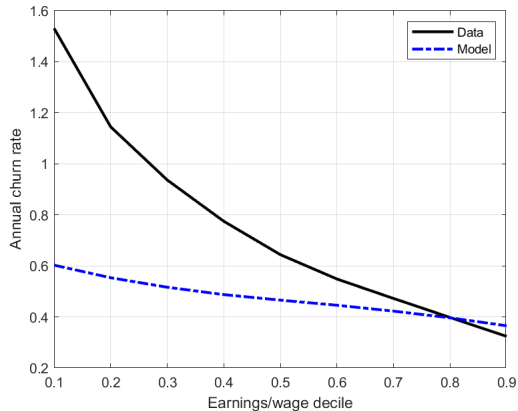


Calibrated churn ladder: Only J2J

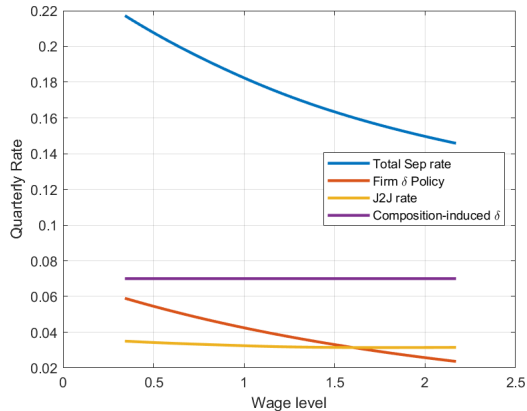


- Considerably less decline in churn over job transitions and less dispersion at firm level.

Calibrated churn ladder: Firm elasticity but No Worker Heterogeneity



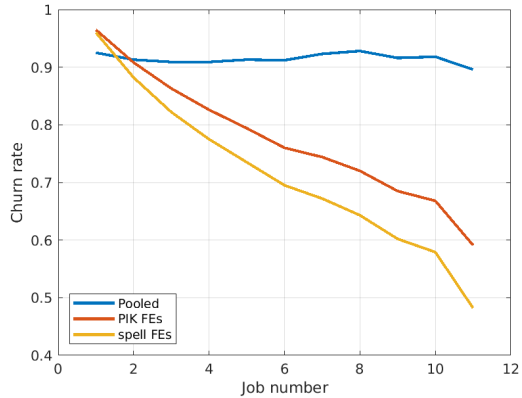
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Conclusions

- Workers move down the churn ladder and up the wage ladder.
- Huge cross-sectional variation in churn in the data.
- Wage-turnover tradeoff from job ladder models is consistent with data.
- On-the-job search and separations to nonemployment both necessary to match churn.
- Worker heterogeneity in separation probability and selection key to matching data.

Fixed Effects



Measuring the job ladder

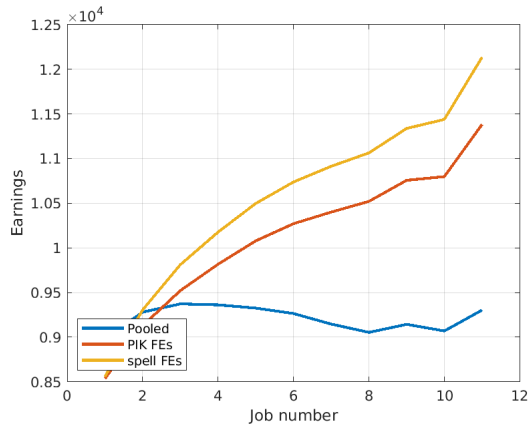
Firm Average Wage

- Index jobs i within continuous employment spell j
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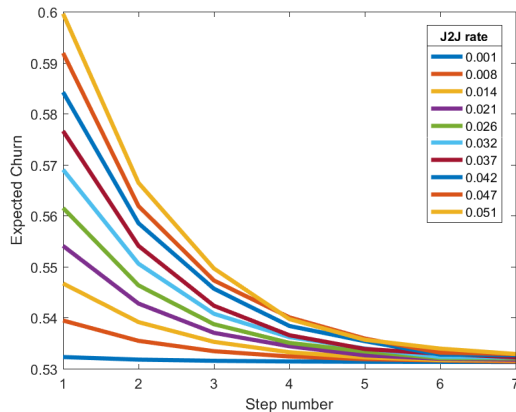
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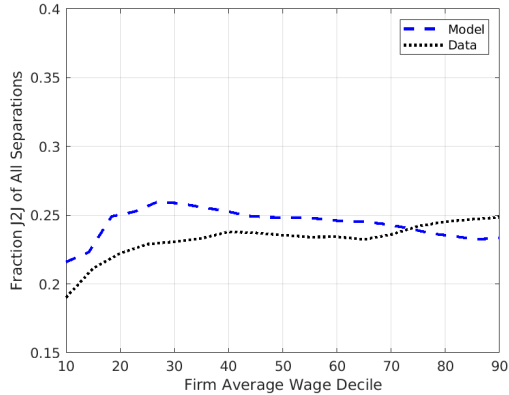


The Churn Ladder as λ_e changes

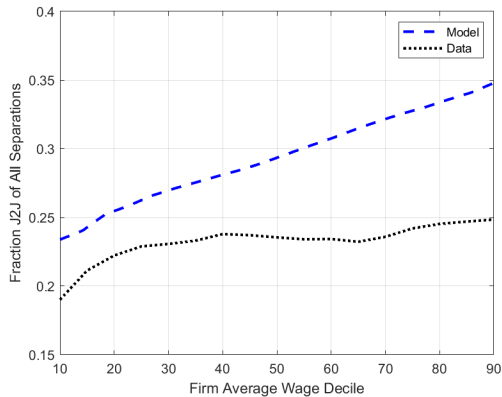


- No heterogeneity in δ

Fraction of all Separations to EE are Constant



Fraction of all Separations to EE are Constant (No Worker Sorting)



Net Growth and Churn

